

SYNOPSIS

AMITY UNIVERSITY UTTAR PRADESH

Amity Institute of Information Technology

NTCC Term Paper, Summer Internship Report

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Project Information

1) Project Duration: 47 Days

Summer Internship Commencement: 18/05/2026

Summer Internship Completion: 03/07/2026

2) Project Topic

RADMAP – RADiation MAPing – GPS Enabled Radiation Mapping Prototype on Raspberry Pi-5 (3D Visualization)

3) Project Objective

To develop a GPS-enabled radiation mapping system capable of real-time radiation detection, environmental data acquisition, and visualization through 2D heatmaps and a 3D Digital Twin. The project aims to enhance radiation monitoring and support future radioactive source localization.

4) Methodology to be Adopted

The project involves integrating a Gamma Spectrometer, GlobalSat BU-353N GPS Receiver, Raspberry Pi-5, and environmental sensors for real-time data acquisition. Radiation data, GPS coordinates, temperature, and humidity are processed through a Python-based backend to generate interactive 2D heatmaps and a 3D Digital Twin visualization with live camera integration and color-graded radiation overlays.

5) Brief Summary of Project

This project focuses on the design and development of a real-time GPS-enabled radiation mapping prototype using Raspberry Pi-5. My work included creating a radiation dataset, developing **2D radiation heatmaps**, integrating GPS telemetry and environmental sensors, validating gamma radiation using **Co-60** and **Cesium** sources, and designing a **3D Digital Twin** for advanced visualization and future source localization. The complete backend system enables live radiation monitoring, geospatial mapping, and intelligent visualization, providing a scalable solution for environmental radiation surveillance and nuclear safety applications.

Signature

(Student)

Signature

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